

Coding & Solution

Date	15 March 2026
Team ID	SWTID-2026-5339
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

- 1 Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- 2 The solution should address the core problem statement and implement the key functional requirements.
- 3 Include all source files, configuration files, and setup instructions needed to run the application.
- 4 Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	(Paste your GitHub Repository Link here)
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	FastAPI, Jinja2 Templates, Google Gemini API, Uvicorn
Key Features Implemented	• AI-powered Question Answering (Q&A) • Topic Explanation Module • Text Summarization • Automatic Quiz Generation (MCQs) • Personalized Learning Path Recommendation • REST API Integration • Responsive Web Interface
Pending / Incomplete Features	• User Authentication and Login • User Progress Tracking • Database Integration for User History • Cloud Deployment
Setup / Run Instructions	1 Install Python 3.10 or above. 2 Install dependencies using <code>pip install -r requirements.txt</code>. 3 Configure the Google Gemini API Key. 4 Run the application using <code>uvicorn main:app --reload</code>. 5 Open <code>http://127.0.0.1:8000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (or "No" if you have not uploaded it to GitHub yet)

Additional Notes / Comments:

The project follows a modular architecture, with separate Python modules for each AI functionality, including Question Answering, Explanation, Summarization, Quiz Generation, and Learning Path Recommendation.

FastAPI is used to develop RESTful APIs, while Google Gemini 1.5 Pro powers the AI features.

The frontend is built using HTML, CSS, and Jinja2 templates, providing an interactive and responsive user interface.

The application has been functionally tested, and all core features operate successfully with proper error handling and JSON-based API communication.